

BUG BOUNTY FIELD GUIDE

Google Dorking

Advanced Search Operators, Sensitive Data Dorks
& Target Reconnaissance for Bug Bounty Hunters

- Search Engine OSINT
- 2025 Edition

Table of Contents

1. Introduction to Google Dorking

2. Core Search Operators

2.1 Basic Operators

2.2 Advanced Operators

2.3 Combining Operators

3. Subdomain Enumeration

4. Sensitive File Discovery

5. Login & Admin Panel Discovery

6. API Keys & Secrets Exposure

7. Database & Backup Exposure

8. Cloud & DevOps Exposure

9. Error Messages & Debug Info

10. Person & Employee Information

11. One-Liners & Automation

12. Best Practices & Legal

1. Introduction to Google Dorking

Google Dorking — also known as Google Hacking — is the technique of using advanced search operators to find sensitive information, hidden pages, and security vulnerabilities that are not intended to be publicly accessible. For bug bounty hunters, it is one of the most powerful **passive reconnaissance** techniques available. 🕵️

Every day, Google crawls and indexes billions of web pages. This includes admin panels, configuration files, exposed databases, API keys, error messages, employee directories, and more. With the right operators, you can filter through this ocean of data to surface high-value findings for your target. 🔍

Why Google Dorking Matters for Bug Bounty

Completely Passive: No packets sent to the target. **Free:** Zero cost, unlimited searches. **Historical:** Finds cached pages that may no longer be live. **Deep:** Indexes content other tools cannot reach. **PII Goldmine:** Surfaces accidentally exposed credentials, keys, and personal data.

Legal Warning

Google Dorking itself is legal — you are only querying publicly indexed data. However, **accessing or downloading sensitive files you discover may be illegal** under laws like CFAA and Computer Misuse Act. Only report findings to authorized bug bounty programs. Never exploit discovered credentials or access restricted systems without permission.

2. Core Search Operators

2.1 Basic Operators

BASICS

Table 1 Basic Google Operators

Operator	Description	Example
"exact phrase"	Exact phrase match	"internal use only"
+word	Include word (must exist)	+admin site:target.com

-word	Exclude word	site:target.com -www
OR /	Either term	admin OR administrator site:target.com
*	Wildcard	target.com/*.php
..	Number range	site:target.com filetype:pdf 2020..2025

2.2 Advanced Operators

ADVANCED

Table 2 Advanced Google Operators

Operator	Description	Example
site:	Restrict to domain	site:target.com
inurl:	Term in URL	inurl:admin site:target.com
intitle:	Term in page title	intitle:"index of" site:target.com
intext:	Term in page body	intext:"password" site:target.com
filetype:	Specific file extension	filetype:env site:target.com
ext:	Alias for filetype	ext:sql site:target.com
cache:	Cached version of page	cache:target.com/admin
related:	Related sites	related:target.com
link:	Pages linking to URL	link:target.com
define:	Word definitions	define:api
allintitle:	All terms in title	allintitle:admin panel login
allinurl:	All terms in URL	allinurl:admin panel
allintext:	All terms in body	allintext:username password
before:	Before a date	site:target.com before:2024-01-01

after:

After a date

site:target.com after:2024-01-01

2.3 Combining Operators

TECHNIQUES

Operator Combination Rules

Combine operators for precision. Use `site:` as your base filter, then layer `inurl:`, `intitle:`, `filetype:`, and `intext:` to drill down. Use `-` to exclude noise (e.g., `-site:github.com` to remove GitHub mirrors). Use `before:` and `after:` to find time-sensitive leaks.

COMBINATION EXAMPLES

```
# Exclude common subdomains to find hidden ones
site:*.target.com -www -mail -ftp -docs -blog

# Find admin panels on specific TLDs
inurl:admin site:target.com OR site:target.co.uk OR site:target.io

# PDF files containing "internal" in the last 2 years
site:target.com filetype:pdf "internal" after:2023-01-01

# Combine site + inurl + intitle for surgical precision
site:target.com inurl:api intitle:"documentation"
```

3. Subdomain Enumeration

Google indexes subdomains as they are crawled. Using wildcard `site:` operators, you can discover subdomains that may not appear in any other tool. 🌐

SUBDOMAIN Basic Subdomain Enumeration

Find all indexed subdomains for a target domain.

```
site:*.target.com
```

SUBDOMAIN Exclude Known Subdomains

Filter out www, mail, blog, docs to find hidden/secondary subdomains.

```
site:*.target.com -www -mail -ftp -blog -docs -support -help
```

SUBDOMAIN Deep Subdomain Search

Find third-level and deeper subdomains that are often staging/dev.

```
site:*.*.target.com
```

SUBDOMAIN Dev/Staging Subdomains

Common naming patterns for development and staging environments.

```
site:*dev*.target.com OR site:*staging*.target.com OR site:*test*.target.com OR  
site:*uat*.target.com OR site:*qa*.target.com
```

SUBDOMAIN API Subdomains

Find API endpoints and developer portals.

```
site:*api*.target.com OR site:*graphql*.target.com OR site:*swagger*.target.com OR  
site:*docs*.target.com
```

SUBDOMAIN Portal/Internal Subdomains

Internal portals, employee logins, and partner access points.

```
site:*portal*.target.com OR site:*internal*.target.com OR  
site:*employee*.target.com OR site:*partner*.target.com OR site:*vpn*.target.com
```

4. Sensitive File Discovery

Files uploaded to websites — backups, configs, documents — often contain sensitive data. Google indexes these if they are not protected by robots.txt or authentication. 📁

CONFIG Environment Files (.env)

.env files contain database credentials, API keys, secrets, and app config.

```
site:target.com filetype:env "DB_PASSWORD" OR "APP_KEY" OR "API_SECRET"
```

CONFIG Config Files (.yaml, .yml, .json, .xml)

Configuration files may expose endpoints, credentials, and internal paths.

```
site:target.com (filetype:yaml OR filetype:yml OR filetype:json OR filetype:xml) ("password" OR "secret" OR "api_key" OR "token")
```

BACKUP Database Backups (.sql, .dump, .bak)

Database dumps expose full table structures, user data, and credentials.

```
site:target.com (filetype:sql OR filetype:dump OR filetype:bak OR inurl:backup OR inurl:dump) ("CREATE TABLE" OR "INSERT INTO" OR "dump")
```

DOCUMENT Internal Documents (.pdf, .docx, .xlsx)

PDFs, Word docs, and spreadsheets marked "internal" or "confidential".

```
site:target.com (filetype:pdf OR filetype:docx OR filetype:xlsx) ("confidential" OR "internal use only" OR "proprietary" OR "do not distribute")
```

SOURCE Git Repository Exposure

Exposed .git directories allow full source code reconstruction with git-dumper.

```
inurl:"/.git" site:target.com "Index of"
```

LOG Log Files (.log)

Log files may contain error traces, session IDs, IPs, paths, and tokens.

```
site:target.com filetype:log ("error" OR "warning" OR "debug" OR "token" OR "session")
```

5. Login & Admin Panel Discovery

Admin panels are high-value targets. When exposed without proper access controls or running outdated software, they can lead to full system compromise. 🗝️

ADMIN Generic Admin Panels

Find all login and admin pages on the target domain.

```
site:target.com (inurl:admin OR inurl:login OR inurl:signin OR inurl:dashboard OR inurl:panel OR inurl:cp OR inurl:backend)
```

ADMIN CMS Admin Pages

Content Management System admin endpoints — WordPress, Joomla, Drupal.

```
site:target.com (inurl:wp-admin OR inurl:administrator OR inurl:admin.php OR inurl:admin/login OR inurl:adminpanel OR inurl:admin_area)
```

ADMIN phpMyAdmin & Database Tools

Database administration interfaces with potential weak credentials.

```
site:target.com (inurl:phpmyadmin OR inurl:phpMyAdmin OR inurl:pma OR inurl:myadmin OR inurl:mysql)
```


ADMIN WebMail & CPanel

Webmail interfaces and hosting control panels.

```
site:target.com (inurl:webmail OR inurl:cpanel OR inurl:whm OR inurl:roundcube OR inurl:squirrelmail)
```

ADMIN Intitle Admin Panels

Use page titles to find admin interfaces regardless of URL structure.

```
site:target.com (intitle:"Admin Panel" OR intitle:"Admin Login" OR intitle:"Administrator" OR intitle:"Control Panel" OR intitle:"Dashboard")
```

ADMIN Forgot Password Pages

Password reset pages can reveal usernames and sometimes be abused for account takeover.

```
site:target.com (inurl:forgot-password OR inurl:reset-password OR inurl:recover OR inurl:forgotpassword)
```

6. API Keys & Secrets Exposure

Developers frequently hardcode API keys, tokens, and secrets in JavaScript files, HTML, and public repositories. Google indexes these when they appear on public pages. 🗝️

SECRET AWS Access Keys

AWS access key IDs starting with AKIA or ASIA exposed in source code.

```
site:target.com "AKIA" OR "ASIA" "aws_access_key_id" OR "aws_secret_access_key"
```

SECRET Google API Keys

Google Maps, reCAPTCHA, Firebase, and Cloud API keys.

```
site:target.com "AIza" ("googleapis" OR "firebase" OR "recaptcha" OR "maps.googleapis")
```

SECRET Stripe Keys

Stripe publishable and secret keys leaked in frontend code.

```
site:target.com ("pk_live_" OR "sk_live_" OR "pk_test_" OR "sk_test_") "stripe"
```

SECRET Slack & Discord Webhooks

Webhook URLs exposed in source code allowing unauthorized message injection.

```
site:target.com ("hooks.slack.com" OR "discord.com/api/webhooks" OR "discordapp.com/api/webhooks")
```

SECRET Private Keys

RSA, EC, and OpenSSH private keys accidentally exposed in web-accessible files.

```
site:target.com ("BEGIN RSA PRIVATE KEY" OR "BEGIN OPENSSH PRIVATE KEY" OR "BEGIN EC PRIVATE KEY" OR "BEGIN DSA PRIVATE KEY")
```

SECRET JWT Secrets & Tokens

Hardcoded JWT signing secrets or bearer tokens in JavaScript and configs.

```
site:target.com ("jwt_secret" OR "jwt_key" OR "Bearer eyJ" OR "HS256" OR "token_secret")
```

Firebase & Cloud Configs

site:target.com "firebaseio.com" OR "appspot.com" OR "cloudfunctions.net" — Exposed mobile app configs often contain database URLs, API keys, and cloud function endpoints.

site:target.com intitle:"index of" "config.json" — Directory listings showing config files with embedded secrets.

7. Database & Backup Exposure

Database dumps and backup files are among the highest-severity findings. They may contain full user tables, password hashes, PII, and internal system data. 🚩

DATABASE SQL Dumps

MySQL, PostgreSQL, and SQLite dumps with full table data and inserts.

```
site:target.com (filetype:sql OR ext:sql) ("CREATE TABLE" OR "INSERT INTO" OR "dump" OR "backup")
```

DATABASE SQLite Databases (.db, .sqlite, .sqlite3)

SQLite database files are complete databases — downloadable and queryable locally.

```
site:target.com (filetype:db OR filetype:sqlite OR filetype:sqlite3 OR inurl:.db) ("SQLite format" OR "sqlite_")
```

BACKUP Backup Archives (.zip, .tar.gz, .rar, .7z)

Compressed backup files may contain entire website source code and databases.

```
site:target.com (filetype:zip OR filetype:tar.gz OR filetype:rar OR filetype:7z OR filetype:gzip) ("backup" OR "site" OR "target" OR "dump" OR "archive")
```

BACKUP WordPress Backups

WordPress backup plugins often store backups in predictable locations.

```
site:target.com (inurl:wp-content OR inurl:wp-content/uploads) (filetype:zip OR filetype:sql OR "backup" OR "wp-backup")
```

BACKUP S3 Bucket Listings

Misconfigured S3 buckets with directory listing enabled.

```
site:s3.amazonaws.com "target.com" OR "ListBucketResult" OR site:*.s3.amazonaws.com "target"
```

BACKUP Old / Backup Domains

Legacy domains and backup copies of the main site.

```
site:*backup.target.com OR site:*old.target.com OR site:*legacy.target.com OR  
site:*archive.target.com
```

8. Cloud & DevOps Exposure

Cloud misconfigurations and exposed DevOps tools are common bug bounty findings. These dorks target CI/CD platforms, container registries, and cloud assets. ☁️

CLOUD Docker Registry

Unauthenticated Docker registries leaking container images and source.

```
site:target.com intitle:"Docker Registry" OR inurl:v2/_catalog
```

CLOUD Jenkins Instances

Exposed Jenkins CI/CD servers with potential script console access.

```
site:target.com intitle:"Dashboard [Jenkins]" OR inurl:jenkins
```

CLOUD Kubernetes Dashboards

Exposed K8s dashboards allowing cluster enumeration.

```
site:target.com intitle:"Kubernetes Dashboard" OR  
inurl:/api/v1/namespaces/kubernetes-dashboard
```

CLOUD Grafana / Prometheus

Monitoring dashboards exposed without authentication.

```
site:target.com (intitle:"Grafana" OR intitle:"Prometheus") OR inurl:/graph OR  
inurl:/metrics
```

CLOUD **Swagger / API Documentation**

Exposed API docs revealing endpoints, parameters, and sometimes test tokens.

```
site:target.com (intitle:"Swagger UI" OR intitle:"API Documentation" OR  
inurl:swagger-ui OR inurl:api-docs OR inurl:openapi)
```

CLOUD **GraphQL Playground**

GraphQL introspection enabled allows full schema discovery.

```
site:target.com (intitle:"GraphQL Playground" OR inurl:/graphql OR  
inurl:/playground OR intitle:"GraphiQL")
```

9. Error Messages & Debug Info

Error pages and debug output reveal technology stacks, file paths, database structures, and sometimes credentials. They are excellent starting points for further testing. ⚠️

DEBUG **PHP Error Messages**

PHP warnings and fatal errors revealing file paths and code structure.

```
site:target.com ("Fatal error:" OR "Warning:" OR "Parse error:" OR "Notice:") "on  
line"
```

DEBUG **MySQL Error Messages**

MySQL errors indicating SQL injection points and database structure.

```
site:target.com ("You have an error in your SQL syntax" OR "mysql_fetch" OR "ORA-"  
OR "PostgreSQL" OR "supplied argument is not a valid MySQL result")
```

DEBUG Stack Traces

Full stack traces revealing frameworks, libraries, and internal paths.

```
site:target.com ("Traceback" OR "Stack trace" OR "at java.lang" OR "at org.springframework" OR "at django.core")
```

DEBUG Debug & Trace Pages

Debug mode pages with environment variables and config dumps.

```
site:target.com (inurl:debug OR inurl:trace OR inurl:phpinfo OR inurl:/.env OR inurl:server-status OR intitle:"phpinfo()")
```

DEBUG Apache & Nginx Status Pages

Server status pages revealing active connections, virtual hosts, and paths.

```
site:target.com (inurl:server-status OR inurl:status OR inurl:nginx_status OR intitle:"Apache Status" OR intitle:"Server Status")
```

DEBUG Directory Listings

Open directory indexes exposing files, backups, and source code.

```
site:target.com intitle:"Index of" "Parent Directory" (".git" OR ".env" OR "backup" OR "config" OR "uploads")
```

10. Person & Employee Information

Finding employee information helps build social engineering profiles, identify key personnel for spear-phishing simulations, and discover exposed contact databases. 👤

PEOPLE Employee Directory Pages

Pages listing employees, their roles, and contact information.

```
site:target.com ("employee directory" OR "staff directory" OR "team" OR "our people" OR "meet the team" OR "contact us")
```

PEOPLE Email Pattern Discovery

Find email addresses to determine corporate email format.

```
site:target.com "@target.com" ("email" OR "contact" OR "reach out" OR "mailto:")
```

PEOPLE LinkedIn Profiles of Employees

LinkedIn pages of target company employees for OSINT profiling.

```
site:linkedin.com/in "target.com" OR "Target Company" ("engineer" OR "developer" OR "security" OR "CTO" OR "CISO")
```

PEOPLE Resume & CV Exposure

Employee resumes uploaded to the site may reveal internal projects and technologies.

```
site:target.com (filetype:pdf OR filetype:doc OR filetype:docx) ("resume" OR "CV" OR "curriculum vitae")
```

PEOPLE Email Distribution Lists

Public mailing list archives with email addresses and internal discussions.

```
site:target.com ("mailing list" OR "mailman" OR "listserv" OR "google groups" OR "forum")
```

PEOPLE Org Chart & Internal Docs

Organizational charts and internal planning documents.

```
site:target.com (filetype:pdf OR filetype:pptx) ("org chart" OR "organization chart" OR "internal" OR "all-hands" OR "quarterly")
```

11. One-Liners & Automation

Chain Google dorks with other tools to build automated reconnaissance pipelines. Here are ready-to-use one-liners. 

GOOP ONE-LINERS

```
# Install goop: pip install goop

# Extract all URLs from Google dork results
goop -t target.com -s "site:*.target.com" -o subdomains.txt

# Find admin panels
goop -t target.com -s "inurl:admin site:target.com" -o admin_urls.txt

# Find .env files
goop -t target.com -s 'filetype:env "DB_PASSWORD" site:target.com' -o env_files.txt

# Multi-dork batch search
goop -t target.com -s "site:target.com filetype:sql" -o sql_files.txt
goop -t target.com -s "site:target.com filetype:pdf confidential" -o internal_pdfs.txt
```

BATCH DORK AUTOMATION SCRIPT


```
#!/bin/bash
# google_dork_recon.sh - Batch Google dorking for bug bounty

TARGET="target.com"
OUTPUT_DIR="recon_${TARGET}"
mkdir -p "$OUTPUT_DIR"

echo "[*] Enumerating subdomains..."
goop -t "$TARGET" -s "site:*.${TARGET} -www -mail" | tee "$OUTPUT_DIR/subdomains.txt"

echo "[*] Finding login/admin pages..."
goop -t "$TARGET" -s "site:${TARGET} (inurl:admin OR inurl:login)" | tee "$OUTPUT_DIR/admin_panels.txt"

echo "[*] Finding .env files..."
goop -t "$TARGET" -s "site:${TARGET} filetype:env" | tee "$OUTPUT_DIR/env_files.txt"

echo "[*] Finding backup files..."
goop -t "$TARGET" -s "site:${TARGET} (filetype:bak OR filetype:zip OR filetype:sql backup)" | tee "$OUTPUT_DIR/backups.txt"

echo "[*] Finding API docs..."
goop -t "$TARGET" -s "site:${TARGET} (swagger OR api-docs OR graphql OR openapi)" | tee "$OUTPUT_DIR/api_docs.txt"

echo "[*] Finding sensitive docs..."
goop -t "$TARGET" -s "site:${TARGET} filetype:pdf (confidential OR internal)" | tee "$OUTPUT_DIR/sensitive_pdfs.txt"

echo "[*] Done! Results in $OUTPUT_DIR/"
```

Google Dorking Best Practices

Use Tor/VPN: Google may rate-limit or CAPTCHA after many searches. **Add delays:** Wait 2-5 seconds between queries to avoid blocks. **Use goop:** Automates result extraction without browser interaction. **Combine with httpx:** Validate found URLs are still live. **Check cache:** Use `cache:` operator for pages that may have been removed. **Document queries:** Always note the exact dork used for reproducibility in reports.

12. Best Practices & Legal

Validation Checklist

Table 3 Before Reporting

Step	Action	Why
1	Verify finding is still live (not just cached)	Google cache may show fixed issues
2	Confirm asset belongs to target	False positives waste triager time
3	Check if in program scope	Out-of-scope reports are rejected
4	Do not download sensitive files	Downloading PII may be illegal
5	Document exact dork and URL found	Reproducibility strengthens report
6	Include screenshot with timestamp	Proves finding at time of discovery

Report Template

Google Dork Finding Report Template

Title: [P1] Exposed .env File Leaking Database Credentials via Google Search

Description: A .env configuration file containing database credentials and API keys is publicly accessible and indexed by Google.

Google Dork Used: `site:target.com filetype:env "DB_PASSWORD"`

URL Found: `https://staging.target.com/.env`

Impact: The .env file contains DB_HOST, DB_PASSWORD, APP_KEY, and MAIL_PASSWORD. An attacker can use these credentials to access the production database, read sensitive customer data, and potentially achieve remote code execution.

Remediation: Remove the .env file from the web root. Add .env to .gitignore and .htaccess deny rules. Rotate all exposed credentials immediately. Review server configuration to prevent future exposure.

Legal & Ethical Guidelines

Querying Google is legal. You are only searching publicly indexed data. **Do not access restricted files.** If you find a .env or .sql file, do not download or open it — just report the URL. **Only report to authorized programs.** Ensure the target has an active bug bounty program. **Never use discovered credentials.** Using exposed credentials is illegal even if they were found passively. **Follow responsible disclosure.** Allow reasonable time for remediation before public disclosure.

Recommended Resources

- **Google Hacking Database (GHDB)** — exploit-db.com/google-hacking-database
- **goop** — github.com/six2dez/goop (Google dorking automation)
- **theHarvester** — laramies/theHarvester (Email/subdomain OSINT)
- **pagodo** — github.com/opsdisk/pagodo (Passive Google dorking)
- **dorks.fyi** — Curated Google dorks for bug bounty
- **Bug Bounty Bootcamp** — Vickie Li, No Starch Press